

Data Structure and Algorithms

BCA — Semester 3 — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Explain algorithm to convert an infix expression to postfix using stack? Use this algorithm to convert $(A+B)*C-D$ to postfix.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Explain merge sort along with its time complexity. Use this algorithm to sort array of numbers given below: 25, 37, 48, 25, 23, 17, 31, 45, 7, 21, 15, 8, 11

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is shortest path algorithm? Use Dijkstra's algorithm to find shortest path between the vertices of a and z in the graph given below.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is data Structure? Explain an array as an abstract data type.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain big oh(O) notation with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define priority queue. How do you implement priority queue? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define recursion. Explain Tower of Hanoi algorithm in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How can you implement queue using linked list? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is binary tree? Explain different application of binary tree.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain sequential search. How is it different from binary search?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define hashing. Explain quadratic probing with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What is graph traversal? Explain breadth first search.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.